

csharp

```
# Typ rekord (record)
```

```
(1,3): error CS1024: Preprocessor directive expected
```

csharp

```
## Úvod a motivácia
```

```
(1,1): error CS1024: Preprocessor directive expected
```

csharp

```
### Príklad - trieda Person
```

```
(1,1): error CS1024: Preprocessor directive expected
```

csharp

```
Máme definovanú nasledovnú triedu a dva objekty s rovnakými hodnotami:
```

```
(1,17): error CS1002: ; expected  
(1,35): error CS1002: ; expected  
(1,41): error CS1002: ; expected  
(1,51): error CS1002: ; expected  
(1,70): error CS1003: Syntax error, ',' expected  
(1,71): error CS1002: ; expected
```

csharp

```
class Person  
{  
    public string FirstName { get; }
```

```
public string LastName { get; }

public Person(string firstName, string lastName)
{
    FirstName = firstName;
    LastName = lastName;
}

var person1 = new Person("Jan", "Mrkvicka");
var person2 = new Person("Jan", "Mrkvicka");
```

csharp

Aké hodnoty vrátia nasledujúce výrazy?

```
(1,13): error CS1002: ; expected
(1,32): error CS1002: ; expected
(1,39): error CS1733: Expected expression
(1,39): error CS1003: Syntax error, ':' expected
(1,39): error CS1733: Expected expression
```

csharp

```
person1.Equals(person2)
```

False

csharp

```
object.Equals(person1, person2)
```

False

csharp

```
person1 == person2
```

False

csharp

```
person1.ToString()
```

Submission#0+Person

csharp

```
Console.WriteLine(person2)
```

Submission#0+Person

csharp

Ako musíme upraviť triedu, aby prvé tri výrazy vracali hodnotu **True** a posledné dve

(1,12): error CS1002: ; expected
(1,28): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement

```
(1,32): error CS1002: ; expected
(1,41): error CS1002: ; expected
(1,56): error CS1002: ; expected
(1,73): error CS1002: ; expected
(1,86): error CS1002: ; expected
(1,93): error CS1002: ; expected
(1,93): error CS7017: Member definition, statement, or end-of-file expected
(1,118): error CS1002: ; expected
(1,126): error CS1002: ; expected
(1,177): error CS1525: Invalid expression term '?'
(1,178): error CS1733: Expected expression
(1,178): error CS1003: Syntax error, ':' expected
(1,178): error CS1733: Expected expression
```

csharp

```
### Příklad - upravená trieda Person
```

```
(1,1): error CS1024: Preprocessor directive expected
```

csharp

```
class Person
{
    public string FirstName { get; }
    public string LastName { get; }

    public Person(string firstName, string lastName)
    {
        FirstName = firstName;
        LastName = lastName;
    }

    public override bool Equals(object obj)
    {
        return obj is Person person &&
            FirstName == person.FirstName &&
            LastName == person.LastName;
    }

    public override int GetHashCode()
    {
```

```

        return GetHashCode.Combine(FirstName, LastName);
    }

    public static bool operator ==(Person left, Person right)
    {
        return EqualityComparer<Person>.Default.Equals(left, right);
    }

    public static bool operator !=(Person left, Person right)
    {
        return !(left == right);
    }

    public override string ToString()
    {
        return $"Person {{ FirstName = {FirstName}, LastName = {LastName} }}";
        //return $"{nameof(Person)} {{ {nameof(FirstName)} = {FirstName}, {nameof(La
    }
}

var person1 = new Person("Jan", "Mrkvicka");
var person2 = new Person("Jan", "Mrkvicka");

Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(person1 == person2);
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);

```

True

True

True

Person { FirstName = Jan, LastName = Mrkvicka }

```
Person { FirstName = Jan, LastName = Mrkvicka }
```

csharp

Celkom dlhá trieda. Môžeme ju trošku zjednodušiť a skrátiť?

```
(1,13): error CS1002: ; expected
(1,31): error CS1002: ; expected
(1,50): error CS1002: ; expected
(1,59): error CS1003: Syntax error, ',' expected
(1,60): error CS1002: ; expected
```

csharp

Príklad - upravená a zjednodušená trieda Person

```
(1,1): error CS1024: Preprocessor directive expected
```

csharp

```
class Person
{
    public string FirstName { get; }
    public string LastName { get; }

    public Person(string firstName, string lastName) => (FirstName, LastName) = (firstName, lastName);

    public override bool Equals(object obj) => obj is Person person &&
        FirstName == person.FirstName &&
        LastName == person.LastName;

    public override int GetHashCode() => GetHashCode.Combine(FirstName, LastName);

    public static bool operator ==(Person left, Person right) => EqualityComparer<Person>.Default.Equals(left, right);
}
```

```

    public static bool operator ≠(Person left, Person right) ⇒ !(left == right);

    public override string ToString() ⇒ $"{nameof(Person)} {{ {nameof(FirstName)}} =
}

var person1 = new Person("Jan", "Mrkvicka");
var person2 = new Person("Jan", "Mrkvicka");

Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(person1 == person2);
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);

```

True

True

True

Person { FirstName = Jan, LastName = Mrkvicka }

Person { FirstName = Jan, LastName = Mrkvicka }

csharp

Stále je celkom dlhá. Neexistuje niečo lepšie?

(1,10): error CS1002: ; expected
(1,34): error CS1003: Syntax error, '(' expected

```
(1,46): error CS1003: Syntax error, ',' expected
(1,47): error CS1026: ) expected
(1,47): error CS1002: ; expected
```

csharp

```
### Príklad - štruktúra Person
```

```
(1,1): error CS1024: Preprocessor directive expected
```

csharp

```
Ak by sme nepotrebovali operátor ==, môžeme deklarovať Person ako štruktúru:
```

```
(1,7): error CS1002: ; expected
(1,25): error CS1002: ; expected
(1,36): error CS1525: Invalid expression term ','
(1,36): error CS1002: ; expected
(1,36): error CS7017: Member definition, statement, or end-of-file expected
(1,56): error CS1002: ; expected
(1,67): error CS1002: ; expected
(1,77): error CS1733: Expected expression
```

csharp

```
struct Person
{
    public string FirstName { get; }
    public string LastName { get; }

    public Person(string firstName, string lastName) => (FirstName, LastName) = (firstName, lastName);

    //public static bool operator ==(Person left, Person right) => EqualityComparer<Person>.Default.Equals(left, right);

    //public static bool operator !=(Person left, Person right) => !(left == right);

    public override string ToString() => $"{nameof(Person)} {{ {nameof(FirstName)} = {FirstName}, {nameof(LastName)} = {LastName} }}";
}

var person1 = new Person("Jan", "Mrkvicka");
```

```
var person2 = new Person("Jan", "Mrkvicka");

Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
//Console.WriteLine(person1 == person2);
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);
```

True

True

Person { FirstName = Jan, LastName = Mrkvicka }

Person { FirstName = Jan, LastName = Mrkvicka }

csharp

Neexistuje predsa niečo lepšie?

```
(1,19): error CS1002: ; expected
(1,31): error CS1003: Syntax error, ',' expected
(1,32): error CS1002: ; expected
```

csharp

Príklad - rekord Person

```
(1,1): error CS1024: Preprocessor directive expected
```

Môžeme si definovať pozičný ****rekord****:

```
(1,11): error CS1002: ; expected
(1,29): error CS1002: ; expected
(1,39): error CS1525: Invalid expression term ':'
(1,39): error CS1002: ; expected
(1,39): error CS7017: Member definition, statement, or end-of-file expected
```

```
record Person(string FirstName, string LastName);

var person1 = new Person("Jan", "Mrkvicka");
var person2 = new Person("Jan", "Mrkvicka");

Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(person1 == person2);
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);
```

True

True

True

Person { FirstName = Jan, LastName = Mrkvicka }

Person { FirstName = Jan, LastName = Mrkvicka }

csharp

```
## Čo je to rekord?
```

```
(1,1): error CS1024: Preprocessor directive expected
```

csharp

Typ **record** je **referenčný** typ – špecializovaná trieda, pre ktorú kompilátor vygeneruje:

- pre zisťovanie rovností na základe hodnôt (musia sedieť typy a hodnoty všetkých vlastností)
 - `override bool Equals(object)`
 - `virtual bool Equals(T)` – implementácia rozhrania `System.IEquatable<T>`, kde `T` je typ
 - `operator==`, `operator!=`
- pre zisťovanie hashu – `override int GetHashCode()`
- pre zobrazenie výstupu vo forme reťazca – `override string ToString()`, pričom táto metóda musí byť nadefinovaná
- pre kopírovanie a klonovanie

Ak sa jedná o **pozičný rekord** (positional record), kompilátor navyše vygeneruje:

- **primárny konštruktor**, ktorého parametre sú rovnaké ako pozičné parametre v dekonstruktor
- **verejné inicializačné** (public init-only) **vlastnosti** pre každý parameter priradený
- **metódu** `Deconstruct()` na extrakciu hodnôt vlastností z rekordu. Táto metóda musí byť nadefinovaná

Kompilátor môže vygenerovať aj ďalšie metódy podľa potreby. Ak si definujete vlastnú

Rekord môžeme vytvoriť pomocou:

1. syntaxe s **pozičnými parametrami** (positional parameters),
2. syntaxe so **štandardnými vlastnosťami** (standard properties),
3. kombináciou.

Typ bol zavedený v C# 9.0.

```
(1,53): error CS1002: ; expected
(1,59): error CS1002: ; expected
(1,59): error CS7017: Member definition, statement, or end-of-file expected
(1,71): error CS1002: ; expected
(1,93): error CS1002: ; expected
(1,113): error CS1002: ; expected
(2,7): error CS1002: ; expected
```

(2,27): error CS1002: ; expected
(2,38): error CS1002: ; expected
(2,52): error CS1003: Syntax error, ',' expected
(2,59): error CS1003: Syntax error, ',' expected
(2,64): error CS1003: Syntax error, ',' expected
(2,66): error CS1003: Syntax error, ',' expected
(2,74): error CS1003: Syntax error, ',' expected
(2,83): error CS1003: Syntax error, ',' expected
(2,94): error CS1003: Syntax error, ',' expected
(2,96): error CS1003: Syntax error, ',' expected
(2,103): error CS1002: ; expected
(2,103): error CS7017: Member definition, statement, or end-of-file expected
(3,7): error CS1525: Invalid expression term ''
(3,7): error CS1002: ; expected
(3,7): error CS1056: Unexpected character ``'
(3,35): error CS1001: Identifier expected
(3,36): error CS1002: ; expected
(3,36): error CS1056: Unexpected character ``'
(4,7): error CS1525: Invalid expression term ''
(4,7): error CS1002: ; expected
(4,7): error CS1056: Unexpected character ``'
(4,29): error CS1001: Identifier expected
(4,30): error CS1002: ; expected
(4,30): error CS1056: Unexpected character ``'
(4,48): error CS1002: ; expected
(4,58): error CS1002: ; expected
(4,58): error CS1056: Unexpected character ``'
(4,79): error CS1525: Invalid expression term ''
(4,79): error CS1002: ; expected
(4,79): error CS1056: Unexpected character ``'
(4,80): error CS7017: Member definition, statement, or end-of-file expected
(4,97): error CS1002: ; expected
(4,104): error CS1002: ; expected
(5,7): error CS1525: Invalid expression term ''
(5,7): error CS1002: ; expected
(5,7): error CS1056: Unexpected character ``'
(5,8): error CS1031: Type expected
(5,16): error CS1534: Overloaded binary operator '==' takes two parameters
(5,18): error CS1003: Syntax error, '(' expected
(5,18): error CS1056: Unexpected character ``'
(5,19): error CS1031: Type expected
(5,19): error CS1001: Identifier expected
(5,21): error CS1031: Type expected
(5,21): error CS1001: Identifier expected
(5,21): error CS1056: Unexpected character ``'
(5,32): error CS1056: Unexpected character ``'
(6,3): error CS1003: Syntax error, ',' expected

(6,18): error CS1003: Syntax error, ',' expected
(6,24): error CS1001: Identifier expected
(6,24): error CS1003: Syntax error, ',' expected
(6,26): error CS1056: Unexpected character ''
(6,36): error CS1003: Syntax error, ',' expected
(6,51): error CS1003: Syntax error, ',' expected
(6,52): error CS8124: Tuple must contain at least two elements.
(6,53): error CS1001: Identifier expected
(6,53): error CS1056: Unexpected character ''
(7,3): error CS1003: Syntax error, ',' expected
(7,18): error CS1003: Syntax error, ',' expected
(7,29): error CS1003: Syntax error, ',' expected
(7,43): error CS1003: Syntax error, ',' expected
(7,45): error CS1056: Unexpected character ''
(7,55): error CS1003: Syntax error, ',' expected
(7,70): error CS1003: Syntax error, ',' expected
(7,71): error CS8124: Tuple must contain at least two elements.
(7,72): error CS1001: Identifier expected
(7,72): error CS1056: Unexpected character ''
(7,87): error CS1003: Syntax error, ',' expected
(7,101): error CS1003: Syntax error, ',' expected
(7,123): error CS1003: Syntax error, ',' expected
(7,132): error CS1056: Unexpected character ''
(7,133): error CS1003: Syntax error, ',' expected
(7,145): error CS1001: Identifier expected
(7,145): error CS1003: Syntax error, ',' expected
(7,159): error CS8124: Tuple must contain at least two elements.
(7,160): error CS1001: Identifier expected
(7,160): error CS1056: Unexpected character ''
(7,172): error CS1003: Syntax error, ',' expected
(7,187): error CS1003: Syntax error, ',' expected
(7,197): error CS1003: Syntax error, ',' expected
(8,3): error CS1003: Syntax error, ',' expected
(8,19): error CS1003: Syntax error, ',' expected
(8,31): error CS1003: Syntax error, ',' expected
(10,7): error CS1003: Syntax error, ',' expected
(10,15): error CS1003: Syntax error, ',' expected
(10,17): error CS1003: Syntax error, ',' expected
(10,31): error CS1003: Syntax error, ',' expected
(10,34): error CS1003: Syntax error, ',' expected
(10,52): error CS8124: Tuple must contain at least two elements.
(10,53): error CS1001: Identifier expected
(10,73): error CS1003: Syntax error, ',' expected
(10,83): error CS1001: Identifier expected
(10,83): error CS1003: Syntax error, ',' expected
(11,5): error CS1003: Syntax error, ',' expected
(11,25): error CS1003: Syntax error, ',' expected

(11,47): error CS1003: Syntax error, ',' expected
(11,58): error CS1003: Syntax error, ',' expected
(11,70): error CS1003: Syntax error, ',' expected
(11,82): error CS1003: Syntax error, ',' expected
(11,101): error CS1031: Type expected
(11,101): error CS1001: Identifier expected
(11,101): error CS1003: Syntax error, ',' expected
(12,5): error CS1003: Syntax error, ',' expected
(12,26): error CS1003: Syntax error, ',' expected
(12,29): error CS1003: Syntax error, ',' expected
(12,30): error CS1031: Type expected
(12,30): error CS8124: Tuple must contain at least two elements.
(12,30): error CS1026:) expected
(12,30): error CS1001: Identifier expected
(12,30): error CS1026:) expected
(12,30): error CS1002: ; expected
(12,41): error CS1519: Invalid token '-' in a member declaration
(12,46): error CS1002: ; expected
(12,46): error CS7017: Member definition, statement, or end-of-file expected
(12,67): error CS1002: ; expected
(12,83): error CS1002: ; expected
(12,119): error CS1003: Syntax error, ',' expected
(12,133): error CS1003: Syntax error, ',' expected
(12,148): error CS1003: Syntax error, ',' expected
(12,154): error CS1003: Syntax error, ',' expected
(12,173): error CS1003: Syntax error, ',' expected
(12,195): error CS1003: Syntax error, ',' expected
(12,205): error CS1001: Identifier expected
(12,206): error CS1002: ; expected
(12,206): error CS7017: Member definition, statement, or end-of-file expected
(13,14): error CS1525: Invalid expression term ''
(13,14): error CS1002: ; expected
(13,14): error CS1056: Unexpected character ''
(13,28): error CS1002: ; expected
(13,28): error CS1056: Unexpected character ''
(13,43): error CS1002: ; expected
(13,61): error CS1002: ; expected
(13,77): error CS1003: Syntax error, '(' expected
(13,87): error CS1003: Syntax error, ',' expected
(13,109): error CS1003: Syntax error, ',' expected
(13,120): error CS1001: Identifier expected
(13,120): error CS1003: Syntax error, ',' expected
(13,122): error CS1003: Syntax error, ',' expected
(13,133): error CS1003: Syntax error, ',' expected
(13,171): error CS1003: Syntax error, ',' expected
(13,181): error CS1001: Identifier expected
(13,192): error CS1003: Syntax error, ',' expected

(13,212): error CS1003: Syntax error, ',' expected
(13,227): error CS1003: Syntax error, ',' expected
(15,17): error CS1003: Syntax error, ',' expected
(15,32): error CS1003: Syntax error, ',' expected
(15,46): error CS1003: Syntax error, ',' expected
(15,59): error CS1003: Syntax error, ',' expected
(15,61): error CS1003: Syntax error, ',' expected
(15,67): error CS1003: Syntax error, ',' expected
(15,86): error CS1003: Syntax error, ',' expected
(15,92): error CS1001: Identifier expected
(15,111): error CS1003: Syntax error, ',' expected
(15,131): error CS1003: Syntax error, ',' expected
(15,146): error CS1003: Syntax error, ',' expected
(15,152): error CS1001: Identifier expected
(15,168): error CS1003: Syntax error, ',' expected
(17,15): error CS1003: Syntax error, ',' expected
(17,31): error CS1003: Syntax error, ',' expected
(18,4): error CS1003: Syntax error, ',' expected
(18,14): error CS1003: Syntax error, ',' expected
(18,16): error CS1003: Syntax error, ',' expected
(18,37): error CS1003: Syntax error, ',' expected
(18,40): error CS1003: Syntax error, ',' expected
(18,62): error CS8124: Tuple must contain at least two elements.
(18,63): error CS1001: Identifier expected
(18,64): error CS1031: Type expected
(18,64): error CS1001: Identifier expected
(18,64): error CS1003: Syntax error, ',' expected
(19,4): error CS1003: Syntax error, ',' expected
(19,15): error CS1003: Syntax error, ',' expected
(19,17): error CS1003: Syntax error, ',' expected
(19,43): error CS1003: Syntax error, ',' expected
(19,46): error CS1003: Syntax error, ',' expected
(19,66): error CS8124: Tuple must contain at least two elements.
(19,67): error CS1001: Identifier expected
(19,68): error CS1031: Type expected
(19,68): error CS1001: Identifier expected
(19,68): error CS1003: Syntax error, ',' expected
(20,4): error CS1003: Syntax error, ',' expected
(22,9): error CS1003: Syntax error, ',' expected
(22,20): error CS1003: Syntax error, ',' expected
(22,21): error CS1040: Preprocessor directives must appear as the first non-whitespace character on a line
(22,27): error CS1001: Identifier expected
(22,27): error CS1026:) expected
(22,27): error CS1002: ; expected

Príklady rekordov

(1,1): error CS1024: Preprocessor directive expected

csharp

Najjednoduchší spôsob vytvorenia rekordu je pomocou **syntaxe s pozičnými parametrami

(1,23): error CS1002: ; expected
(1,42): error CS1002: ; expected
(1,53): error CS1002: ; expected
(1,63): error CS1002: ; expected
(1,75): error CS1002: ; expected
(1,88): error CS1525: Invalid expression term ':'
(1,88): error CS1002: ; expected
(1,88): error CS7017: Member definition, statement, or end-of-file expected

csharp

```
public record Person(string FirstName, string LastName, DateTime Birthdate);  
  
var person = new Person("Ján", "Mrkvička", new DateTime(2000, 12, 31));  
person
```

▼ *Person { FirstName = Ján, LastName = Mrkvička, Birthdate = 12/31/2000 12:00:00 AM }*

FirstName	Ján
LastName	Mrkvička
Birthdate	2000-12-31 00:00:00Z

csharp

Iný spôsob je pomocou ****syntaxe so štandardnými vlastnosťami**** (pozor – konštruktor

```
(1,12): error CS1002: ; expected
(1,23): error CS1002: ; expected
(1,33): error CS1002: ; expected
(1,49): error CS1002: ; expected
(1,85): error CS1026: ) expected
(1,85): error CS1002: ; expected
(1,90): error CS1002: ; expected
(1,104): error CS1002: ; expected
(1,123): error CS1003: Syntax error, ',' expected
(1,126): error CS1002: ; expected
```

csharp

```
public record Person
{
    public string FirstName { get; init; }
    public string LastName { get; init; }
    public DateTime Birthdate { get; init; }
};

// Error CS1729: 'Person' does not contain a constructor that takes 3 arguments
//var person = new Person("Ján", "Mrkvička", new DateTime(2000, 12, 31));

// Môžeme nastaviť jednotlivé vlastnosti v inicializačnej časti:
var person = new Person() { FirstName = "Ján", LastName = "Mrkvička", Birthdate = ne
display(person);

// Ak nenastavíme, všetky vlastnosti budú mať implicitné (default) hodnoty
var person2 = new Person();
display(person2);
```

```
(13,1): error CS0103: The name 'display' does not exist in the current context
(17,1): error CS0103: The name 'display' does not exist in the current context
```

csharp

Môžeme použiť aj ****kombináciu pozičných parametrov s vlastnosťami****:

```
(1,15): error CS1002: ; expected
(1,31): error CS1002: ; expected
(1,52): error CS1002: ; expected
(1,66): error CS1002: ; expected
(1,68): error CS1525: Invalid expression term ':'
(1,68): error CS1002: ; expected
(1,68): error CS7017: Member definition, statement, or end-of-file expected
```

csharp

```
public record Person(string FirstName, string LastName)
{
    public DateTime Birthdate { get; init; }
}

// Error CS1729: 'Person' does not contain a constructor that takes 3 arguments
//var person = new Person("Ján", "Mrkvička", new DateTime(2000, 12, 31));

var person = new Person("Ján", "Mrkvička");
display(person);

var person2 = new Person("Eleonóra", "Mrkvičková") { Birthdate = new DateTime(2000,
display(person2);
```

```
(10,1): error CS0103: The name 'display' does not exist in the current context
(13,1): error CS0103: The name 'display' does not exist in the current context
```

csharp

Ak potrebujeme, môžeme si definovať aj ****konštruktory****:

```
(1,17): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement
(1,24): error CS1002: ; expected
(1,37): error CS1002: ; expected
```

```
(1,56): error CS1525: Invalid expression term ':'  
(1,56): error CS1002: ; expected  
(1,56): error CS7017: Member definition, statement, or end-of-file expected
```

csharp

```
public record Person  
{  
    public string FirstName { get; init; }  
    public string LastName { get; init; }  
    public DateTime Birthdate { get; init; }  
  
    public Person(string firstName, string lastName) => (FirstName, LastName) = (fir  
    public Person(string firstName, string lastName, DateTime birthdate) => (FirstNa  
}  
  
// Error CS1729: 'Person' does not contain a constructor that takes 0 arguments  
//var person = new Person();  
  
var person = new Person("Ján", "Mrkvička");  
display(person);  
  
var person2 = new Person("Eleonóra", "Mrkvičková", new DateTime(2000, 12, 31));  
display(person2);
```

```
(15,1): error CS0103: The name 'display' does not exist in the current context  
(18,1): error CS0103: The name 'display' does not exist in the current context
```

csharp

Môžeme kombinovať, pridať aj ****ďalšie vlastnosti**** a ****konštruktor****:

```
(1,20): error CS1044: Cannot use more than one type in a for, using, fixed, or  
declaration statement  
(1,27): error CS1002: ; expected  
(1,39): error CS1002: ; expected  
(1,68): error CS1525: Invalid expression term ':'  
(1,68): error CS1002: ; expected  
(1,68): error CS7017: Member definition, statement, or end-of-file expected
```

```

public record Person(string FirstName, string LastName, DateTime Birthdate)
{
    public string FullName => $"{FirstName} {LastName}";

    public int Age
    {
        get
        {
            var today = DateTime.Today;
            var age = today.Year - Birthdate.Year;
            if (Birthdate.Date > today.AddYears(-age))
                age--;
            return age;
        }
    }

    public Person(string firstName, string lastName) : this(firstName, lastName, new
}

var person = new Person("Ján", "Mrkvička");
display(person);

var person2 = new Person("Eleonóra", "Mrkvičková", new DateTime(2000, 12, 31));
display(person2);

var person3 = new Person("Fero", "Nový") { Birthdate = DateTime.Now.AddYears(-30) };
display(person3);

```

(21,1): error CS0103: The name 'display' does not exist in the current context
 (24,1): error CS0103: The name 'display' does not exist in the current context
 (27,1): error CS0103: The name 'display' does not exist in the current context

Ak nám nevyhovuje ****modifikátor prístupu**** vlastnosti (dokonca aj pozičného parametra)

```

```csharp
 public record Person(string FirstName, string LastName)
 {
 internal string FirstName { get; init; } = FirstName;
 }

```

...

Pozor, tu je nutné ešte nastaviť vlastnosť na hodnotu získanú z pozičného parametra

Otázky:

- Ak zmeníme modifikátor prístupu na `protected`, bude možné skompilovať kód?
- Ak zmeníme modifikátor prístupu na `protected internal`, bude možné skompilovať kód?
- Ak odstránime inicializáciu vlastnosti `FirstName`, čo bude v tejto vlastnosti?

```
(1,8): error CS1002: ; expected
(1,33): error CS1002: ; expected
(1,64): error CS1003: Syntax error, ',' expected
(1,67): error CS1003: Syntax error, ',' expected
(1,77): error CS1003: Syntax error, ',' expected
(1,87): error CS1002: ; expected
(1,87): error CS7017: Member definition, statement, or end-of-file expected
(1,99): error CS1002: ; expected
(1,119): error CS1001: Identifier expected
(1,119): error CS1056: Unexpected character ''
(1,120): error CS1026:) expected
(1,120): error CS1002: ; expected
(1,128): error CS1519: Invalid token '' in a member declaration
(1,128): error CS1056: Unexpected character ''
(1,129): error CS7017: Member definition, statement, or end-of-file expected
(2,1): error CS1056: Unexpected character ''
(2,2): error CS1056: Unexpected character ''
(2,3): error CS1056: Unexpected character ''
(2,10): error CS1002: ; expected
(7,1): error CS1056: Unexpected character ''
(7,2): error CS1056: Unexpected character ''
(7,3): error CS1056: Unexpected character ''
(8,6): error CS1002: ; expected
(8,6): error CS7017: Member definition, statement, or end-of-file expected
(8,14): error CS1002: ; expected
(8,25): error CS1002: ; expected
(8,44): error CS1002: ; expected
(8,55): error CS1002: ; expected
(8,65): error CS1002: ; expected
(8,86): error CS1056: Unexpected character ''
(8,96): error CS1001: Identifier expected
(8,96): error CS1056: Unexpected character ''
(8,98): error CS1002: ; expected
(8,98): error CS7017: Member definition, statement, or end-of-file expected
(8,106): error CS1002: ; expected
(8,120): error CS1002: ; expected
```

(8,139): error CS1002: ; expected  
(8,147): error CS1002: ; expected  
(8,147): error CS7017: Member definition, statement, or end-of-file expected  
(8,164): error CS1002: ; expected  
(8,175): error CS1002: ; expected  
(8,175): error CS1056: Unexpected character ``'  
(8,182): error CS1519: Invalid token ``' in a member declaration  
(8,182): error CS1056: Unexpected character ``'  
(8,183): error CS7017: Member definition, statement, or end-of-file expected  
(11,6): error CS1002: ; expected  
(11,26): error CS1002: ; expected  
(11,38): error CS1056: Unexpected character ``'  
(11,39): error CS1002: ; expected  
(11,48): error CS1519: Invalid token ``' in a member declaration  
(11,48): error CS1056: Unexpected character ``'  
(11,49): error CS7017: Member definition, statement, or end-of-file expected  
(11,62): error CS1002: ; expected  
(11,77): error CS1003: Syntax error, ',' expected  
(11,78): error CS1002: ; expected  
(12,6): error CS1002: ; expected  
(12,26): error CS1002: ; expected  
(12,38): error CS1056: Unexpected character ``'  
(12,39): error CS1002: ; expected  
(12,57): error CS1519: Invalid token ``' in a member declaration  
(12,57): error CS1056: Unexpected character ``'  
(12,58): error CS7017: Member definition, statement, or end-of-file expected  
(12,71): error CS1002: ; expected  
(12,86): error CS1003: Syntax error, ',' expected  
(12,87): error CS1002: ; expected  
(13,6): error CS1002: ; expected  
(13,31): error CS1002: ; expected  
(13,42): error CS1002: ; expected  
(13,42): error CS1056: Unexpected character ``'  
(13,52): error CS1002: ; expected  
(13,52): error CS1056: Unexpected character ``'  
(13,53): error CS7017: Member definition, statement, or end-of-file expected  
(13,63): error CS1002: ; expected  
(13,71): error CS1002: ; expected  
(13,84): error CS1733: Expected expression  
(13,84): error CS1003: Syntax error, ':' expected  
(13,84): error CS1733: Expected expression

csharp

```
public record Person(string FirstName, string LastName)
{
```

```
 internal string FirstName { get; init; } = FirstName;
 }

 var person = new Person("Ján", "Mrkvička");
 display(person);
 Console.WriteLine($"{person.FirstName} {person.LastName}")
}
```

(7,1): error CS0103: The name 'display' does not exist in the current context

csharp

Implicitne (defaultne) sú rekordy **\*\*nemenné (immutable)\*\***, preto nie je možné priradiť

```
(1,24): error CS1002: ; expected
(1,35): error CS1002: ; expected
(1,58): error CS1525: Invalid expression term ',',
(1,58): error CS1002: ; expected
(1,58): error CS7017: Member definition, statement, or end-of-file expected
(1,70): error CS1002: ; expected
(1,79): error CS1002: ; expected
(1,96): error CS1002: ; expected
(1,130): error CS1002: ; expected
(1,130): error CS1003: Syntax error, 'while' expected
(1,130): error CS1003: Syntax error, '(' expected
(1,130): error CS1525: Invalid expression term ':'
(1,130): error CS1026:) expected
(1,130): error CS1002: ; expected
(1,130): error CS7017: Member definition, statement, or end-of-file expected
```

csharp

```
public record Person(string FirstName, string LastName);

// Error CS8852: Init-only property or indexer 'Person.FirstName' can only be assigned
var person = new Person("Ján", "Mrkvička");
person.FirstName = "Fero";
}
```

(5,1): error CS8852: Init-only property or indexer 'Person.FirstName' can only be

assigned in an object initializer, or on 'this' or 'base' in an instance constructor or an 'init' accessor.

csharp

Ak by sme ale definovali **vlastnosť** s časťou **set** (namiesto init), potom vlast

```
(1,7): error CS1002: ; expected
(1,15): error CS1002: ; expected
(1,42): error CS1002: ; expected
(1,67): error CS1026:) expected
(1,67): error CS1002: ; expected
(1,71): error CS1002: ; expected
(1,71): error CS7017: Member definition, statement, or end-of-file expected
(1,90): error CS1002: ; expected
(1,102): error CS1002: ; expected
(1,110): error CS1733: Expected expression
```

csharp

```
public record Person(string FirstName, string LastName)
{
 public string LastName { get; set; } = LastName;
}

var person = new Person("Eleonóra", "Slobodná");
display(person);

person.LastName = "Vydatá";
display(person);
```

```
(7,1): error CS0103: The name 'display' does not exist in the current context
(10,1): error CS0103: The name 'display' does not exist in the current context
```

csharp

Pozičné vlastnosti sú síce nemenné, avšak **údaje v** ich **referenčných hodnotách**

```
(1,20): error CS1002: ; expected
(1,28): error CS1002: ; expected
(1,35): error CS1002: ; expected
(1,35): error CS7017: Member definition, statement, or end-of-file expected
(1,51): error CS1002: ; expected
(1,74): error CS1002: ; expected
(1,85): error CS1525: Invalid expression term ','
(1,85): error CS1002: ; expected
(1,85): error CS7017: Member definition, statement, or end-of-file expected
(1,96): error CS1002: ; expected
(1,104): error CS1002: ; expected
(1,104): error CS7017: Member definition, statement, or end-of-file expected
(1,111): error CS1002: ; expected
(1,122): error CS1002: ; expected
(1,137): error CS1002: ; expected
(1,145): error CS1002: ; expected
(1,161): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement
(1,169): error CS1002: ; expected
(1,178): error CS1002: ; expected
(1,195): error CS1003: Syntax error, ',' expected
(1,197): error CS1002: ; expected
```

csharp

```
public record Person(string FirstName, string LastName, string[] PhoneNumbers);

var person = new Person("Ján", "Mrkvička", new[] { "+421 944 123 456" });
Console.WriteLine(person.PhoneNumbers[0]);

// Zmeniť celé pole nie je možné, pretože vlastnosť PhoneNumbers je definovaná ako i
// Error CS8852: Init-only property or indexer 'Person.PhoneNumbers' can only be ass
//person.PhoneNumbers = new[] { "+421 944 123 456" };

// Nastaviť hodnotu v poli je možné:
person.PhoneNumbers[0] = "+421 908 666 666";
Console.WriteLine(person.PhoneNumbers[0]);
```

+421 944 123 456

+421 908 666 666

csharp

Pozrime sa na metódy pre rovnosť dvoch inšancií:

```
(1,12): error CS1002: ; expected
(1,22): error CS1002: ; expected
(1,34): error CS1002: ; expected
(1,49): error CS1003: Syntax error, ',' expected
(1,50): error CS1002: ; expected
```

csharp

```
public record Person(string FirstName, string LastName, string[] PhoneNumbers);

var phoneNumbers = new string[2];
var person1 = new Person("Ján", "Mrkvička", phoneNumbers);
var person2 = new Person("Ján", "Mrkvička", phoneNumbers);

display(person1);
display(person2);

Console.WriteLine(person1 == person2);
Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(object.ReferenceEquals(person1, person2));
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);
```

```
(7,1): error CS0103: The name 'display' does not exist in the current context
(8,1): error CS0103: The name 'display' does not exist in the current context
```

csharp

Keď \*\*zmeníme telefónne číslo prvej osoby\*\*, budú sa osoby rovnáť?

```
(1,15): error CS1002: ; expected
(1,31): error CS1002: ; expected
(1,42): error CS1002: ; expected
(1,44): error CS1525: Invalid expression term ',',
(1,44): error CS1002: ; expected
(1,44): error CS7017: Member definition, statement, or end-of-file expected
(1,54): error CS1002: ; expected
(1,66): error CS1003: Syntax error, ',' expected
(1,67): error CS1002: ; expected
```

csharp

```
person1.PhoneNumbers[0] = "+421 908 666 666";

display(person1);
display(person2);

Console.WriteLine(person1 == person2);
Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(object.ReferenceEquals(person1, person2));
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);
```

```
(1,9): error CS1061: 'Person' does not contain a definition for 'PhoneNumbers' and no
accessible extension method 'PhoneNumbers' accepting a first argument of type 'Person'
could be found (are you missing a using directive or an assembly reference?)
(3,1): error CS0103: The name 'display' does not exist in the current context
(4,1): error CS0103: The name 'display' does not exist in the current context
```

csharp

Ak by sme chceli **\*\*porovnať dva rekordy s inými názvami rekordov\*\***, ale s rovnakým p

```
(1,7): error CS1002: ; expected
(1,18): error CS1002: ; expected
(1,29): error CS1002: ; expected
(1,41): error CS1002: ; expected
(1,49): error CS1002: ; expected
```

```
(1,65): error CS1002: ; expected
(1,67): error CS1525: Invalid expression term ',',
(1,67): error CS1002: ; expected
(1,67): error CS7017: Member definition, statement, or end-of-file expected
(1,75): error CS1002: ; expected
(1,91): error CS1002: ; expected
(1,99): error CS1002: ; expected
(1,121): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement
(1,126): error CS1002: ; expected
(1,137): error CS1002: ; expected
(1,144): error CS1733: Expected expression
(1,144): error CS1003: Syntax error, ':' expected
(1,144): error CS1733: Expected expression
```

csharp

```
public record Person(string FirstName, string LastName);
public record Osoba(string FirstName, string LastName);

var person1 = new Osoba("Ján", "Mrkvička");
var person2 = new Person("Ján", "Mrkvička");

// Error CS0019: Operator '==' cannot be applied to operands of type 'Osoba' and 'Pe
//Console.WriteLine(person1 == person2);
Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(object.ReferenceEquals(person1, person2));
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);
```

False

False

False

```
Osoba { FirstName = JĎn, LastName = MrkviĎka }
```

```
Person { FirstName = JĎn, LastName = MrkviĎka }
```

csharp

Ak potrebujete **\*\*zmenit nemeniteľnú (immutable) vlastnosť\*\*** inšancie rekordu, je mo

```
(1,16): error CS1002: ; expected
(1,25): error CS1002: ; expected
(1,49): error CS1002: ; expected
(1,71): error CS1002: ; expected
(1,78): error CS1002: ; expected
(1,78): error CS7017: Member definition, statement, or end-of-file expected
(1,89): error CS1002: ; expected
(1,102): error CS1056: Unexpected character ``
(1,103): error CS1002: ; expected
(1,107): error CS1002: ; expected
(1,107): error CS1056: Unexpected character ``
(1,121): error CS1002: ; expected
(1,126): error CS1001: Identifier expected
(1,143): error CS1002: ; expected
(1,169): error CS1026:) expected
(1,169): error CS1002: ; expected
(1,177): error CS1002: ; expected
(1,177): error CS7017: Member definition, statement, or end-of-file expected
```

csharp

```
public record Person(string FirstName, string LastName)
{
 public string[] PhoneNumbers { get; init; }
}

var person1 = new Person("Jana", "Mrkvičková") { PhoneNumbers = new[] { "+421 944 12
var person2 = person1 with { LastName = "Vydatá" };
```

```

display(person1);
display(person2);

Console.WriteLine(person1 == person2);
Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(object.ReferenceEquals(person1, person2));
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);

```

(9,1): error CS0103: The name 'display' does not exist in the current context  
 (10,1): error CS0103: The name 'display' does not exist in the current context

csharp

Výraz `with` vytvorí novú inštanciu rekordu, ktorá je kópiou existujúcej inštancie r  
 Ak **zmeníme meno** opäť a ešte aj **telefónne číslo druhej osoby**, dostaneme:

(1,7): error CS1002: ; expected  
 (1,7): error CS1056: Unexpected character ``  
 (1,12): error CS1002: ; expected  
 (1,12): error CS1056: Unexpected character ``  
 (1,27): error CS1002: ; expected  
 (1,46): error CS1044: Cannot use more than one type in a for, using, fixed, or  
 declaration statement  
 (1,52): error CS1002: ; expected  
 (1,62): error CS1002: ; expected  
 (1,84): error CS1002: ; expected  
 (1,91): error CS1002: ; expected  
 (1,91): error CS7017: Member definition, statement, or end-of-file expected  
 (1,103): error CS1002: ; expected  
 (1,116): error CS1001: Identifier expected  
 (1,116): error CS1003: Syntax error, '(' expected  
 (1,116): error CS1001: Identifier expected  
 (1,130): error CS1003: Syntax error, ',' expected  
 (1,133): error CS1003: Syntax error, ',' expected  
 (1,134): error CS1031: Type expected  
 (1,143): error CS8124: Tuple must contain at least two elements.  
 (1,143): error CS1026: ) expected  
 (1,147): error CS1003: Syntax error, ',' expected

(1,149): error CS1002: ; expected  
(1,149): error CS7017: Member definition, statement, or end-of-file expected  
(1,163): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement  
(1,166): error CS1002: ; expected  
(1,171): error CS1002: ; expected  
(1,192): error CS1002: ; expected  
(1,210): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement  
(1,219): error CS1002: ; expected  
(1,227): error CS1002: ; expected  
(1,241): error CS1002: ; expected  
(1,256): error CS1003: Syntax error, ',' expected  
(1,261): error CS1003: Syntax error, ',' expected  
(1,274): error CS1002: ; expected  
(1,282): error CS1002: ; expected  
(1,282): error CS1056: Unexpected character ``'  
(1,287): error CS1002: ; expected  
(1,287): error CS1056: Unexpected character ``'  
(1,304): error CS1002: ; expected  
(1,322): error CS1002: ; expected  
(1,333): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement  
(1,339): error CS1002: ; expected  
(1,347): error CS1002: ; expected  
(1,360): error CS1002: ; expected  
(1,360): error CS1056: Unexpected character ``'  
(1,380): error CS1002: ; expected  
(1,380): error CS1513: } expected  
(1,399): error CS1002: ; expected  
(1,399): error CS1513: } expected  
(1,401): error CS8635: Unexpected character sequence '... '  
(1,405): error CS1002: ; expected  
(1,406): error CS1056: Unexpected character ``'  
(1,407): error CS7017: Member definition, statement, or end-of-file expected  
(1,427): error CS1002: ; expected  
(1,431): error CS1002: ; expected  
(1,431): error CS1056: Unexpected character ``'  
(1,436): error CS1002: ; expected  
(1,436): error CS1056: Unexpected character ``'  
(1,444): error CS1002: ; expected  
(1,444): error CS1056: Unexpected character ``'  
(1,448): error CS1002: ; expected  
(1,448): error CS1056: Unexpected character ``'  
(3,14): error CS1002: ; expected  
(3,26): error CS1002: ; expected  
(3,33): error CS1002: ; expected

```
(3,48): error CS1002: ; expected
(3,61): error CS1002: ; expected
(3,68): error CS1525: Invalid expression term ','
(3,68): error CS1002: ; expected
(3,68): error CS7017: Member definition, statement, or end-of-file expected
(3,80): error CS1733: Expected expression
```

csharp

```
person2 = person2 with { LastName = "Kalerábiková" };
person2.PhoneNumbers[0] = null;

display(person1);
display(person2);

Console.WriteLine(person1 == person2);
Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(object.ReferenceEquals(person1, person2));
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);
```

```
(2,9): error CS1061: 'Person' does not contain a definition for 'PhoneNumbers' and no
accessible extension method 'PhoneNumbers' accepting a first argument of type 'Person'
could be found (are you missing a using directive or an assembly reference?)
(4,1): error CS0103: The name 'display' does not exist in the current context
(5,1): error CS0103: The name 'display' does not exist in the current context
(7,19): error CS0019: Operator '==' cannot be applied to operands of type 'Osoba' and
'Person'
```

csharp

Môžeme dokonca vytvoriť **\*\*plytkú kópiu prvej osoby\*\*** bez zmeny:

```
(1,16): error CS1002: ; expected
(1,34): error CS1002: ; expected
(1,46): error CS1002: ; expected
(1,58): error CS1002: ; expected
(1,64): error CS1733: Expected expression
```

```

person2 = person1 with { };

display(person1);
display(person2);

Console.WriteLine(person1 == person2);
Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(object.ReferenceEquals(person1, person2)); // False
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);

```

```

(1,11): error CS0029: Cannot implicitly convert type 'Osoba' to 'Person'
(3,1): error CS0103: The name 'display' does not exist in the current context
(4,1): error CS0103: The name 'display' does not exist in the current context
(6,19): error CS0019: Operator '==' cannot be applied to operands of type 'Osoba' and
'Person'

```

Ak priradíme do premennej hodnotu inej inštancie, dochádza ku **\*\*kópii** referencie na

```

(1,14): error CS1002: ; expected
(1,35): error CS1002: ; expected
(1,35): error CS1003: Syntax error, 'while' expected
(1,35): error CS1003: Syntax error, '(' expected
(1,40): error CS1026:) expected
(1,40): error CS1002: ; expected
(1,49): error CS1002: ; expected
(1,49): error CS7017: Member definition, statement, or end-of-file expected
(1,63): error CS1002: ; expected
(1,71): error CS1002: ; expected
(1,85): error CS1002: ; expected
(1,99): error CS1001: Identifier expected
(1,101): error CS1002: ; expected
(1,120): error CS1002: ; expected
(1,120): error CS7017: Member definition, statement, or end-of-file expected

```

csharp

```
person2 = person1;

display(person1);
display(person2);

Console.WriteLine(person1 == person2);
Console.WriteLine(person1.Equals(person2));
Console.WriteLine(object.Equals(person1, person2));
Console.WriteLine(object.ReferenceEquals(person1, person2)); // True
Console.WriteLine(person1.ToString());
Console.WriteLine(person2);
```

```
(1,11): error CS0029: Cannot implicitly convert type 'Osoba' to 'Person'
(3,1): error CS0103: The name 'display' does not exist in the current context
(4,1): error CS0103: The name 'display' does not exist in the current context
(6,19): error CS0019: Operator '==' cannot be applied to operands of type 'Osoba' and
'Person'
```

csharp

```
Príklady - dedičnosť
```

```
(1,1): error CS1024: Preprocessor directive expected
```

csharp

Rekord môžeme podediť pomocou dvojbodky s uvedením triedy a špecifikovaním hodnôt pr

```
(1,15): error CS1002: ; expected
(1,31): error CS1002: ; expected
(1,43): error CS1002: ; expected
(1,59): error CS1002: ; expected
(1,76): error CS1002: ; expected
(1,94): error CS1002: ; expected
(1,117): error CS1002: ; expected
(1,124): error CS1733: Expected expression
```

csharp

```
public abstract record Person(string FirstName, string LastName);

public record Student(string FirstName, string LastName, int Grade) : Person(FirstName, LastName)

Person student = new Student("Ján", "Mrkvička", 2);
display(student);
Console.WriteLine(student);
```

(6,1): error CS0103: The name 'display' does not exist in the current context

csharp

Ak ešte vytvoríme učiteľa, ktorý dedí od osoby, budú sa inštancie rovnať?

(1,9): error CS1002: ; expected  
(1,28): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement  
(1,34): error CS1002: ; expected  
(1,42): error CS1002: ; expected  
(1,47): error CS1002: ; expected  
(1,47): error CS7017: Member definition, statement, or end-of-file expected  
(1,57): error CS1002: ; expected  
(1,73): error CS1003: Syntax error, ',' expected  
(1,74): error CS1002: ; expected

csharp

```
public record Teacher(string FirstName, string LastName, int Grade) : Person(FirstName, LastName)

Person teacher = new Teacher("Ján", "Mrkvička", 2);
Person student = new Student("Ján", "Mrkvička", 2);
Console.WriteLine(teacher);
Console.WriteLine(student);
Console.WriteLine(teacher == student);

Console.WriteLine();
```

```
Student student2 = new Student("Ján", "Mrkvička", 2);
Console.WriteLine(student2);
Console.WriteLine(student);
Console.WriteLine(student2 == student);
```

(11,1): error CS0246: The type or namespace name 'Student' could not be found (are you missing a using directive or an assembly reference?)  
(4,22): error CS0246: The type or namespace name 'Student' could not be found (are you missing a using directive or an assembly reference?)  
(11,24): error CS0246: The type or namespace name 'Student' could not be found (are you missing a using directive or an assembly reference?)

csharp

```
Príklady - Deconstruct metóda
```

(1,1): error CS1024: Preprocessor directive expected

csharp

Kompilátor generuje metódu `Deconstruct()` pre pozičné rekordy. Vďaka nej môžeme dek

(1,21): error CS1002: ; expected  
(1,28): error CS1002: ; expected  
(1,28): error CS1056: Unexpected character ``'  
(1,42): error CS1002: ; expected  
(1,42): error CS1056: Unexpected character ``'  
(1,56): error CS1002: ; expected  
(1,75): error CS1002: ; expected  
(1,96): error CS1002: ; expected  
(1,111): error CS1002: ; expected  
(1,114): error CS1002: ; expected  
(1,114): error CS7017: Member definition, statement, or end-of-file expected  
(1,119): error CS1002: ; expected  
(1,136): error CS1002: ; expected  
(1,136): error CS1003: Syntax error, 'while' expected  
(1,136): error CS1003: Syntax error, '(' expected  
(1,143): error CS1026: ) expected

```
(1,143): error CS1002: ; expected
(1,153): error CS1056: Unexpected character ''
(1,185): error CS1056: Unexpected character ''
(1,187): error CS1002: ; expected
(1,187): error CS7017: Member definition, statement, or end-of-file expected
```

csharp

```
public record Student(string FirstName, string LastName, int Grade);

var student = new Student("Ján", "Mrkvička", 2);
string first, last;
int number;
(first, last, number) = student;
Console.WriteLine($"{first} {last} {number}");
```

Ján Mrkvička 2

csharp

Môžeme to tiež zjednodušiť s použitím kľúčového slova var:

```
(1,11): error CS1002: ; expected
(1,28): error CS1002: ; expected
(1,39): error CS1002: ; expected
(1,55): error CS1002: ; expected
(1,59): error CS1733: Expected expression
```

csharp

```
public record Student(string FirstName, string LastName, int Grade);

var student = new Student("Ján", "Mrkvička", 2);
var (first, last, number) = student;
Console.WriteLine($"{first} {last} {number}");
```

Ján Mrkvička 2

csharp

Ak nejakú vlastnosť nepotrebuje, môžeme použiť `\_` (\*discard\*) premennú:

```
(1,11): error CS1002: ; expected
(1,36): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement
(1,43): error CS1002: ; expected
(1,50): error CS1002: ; expected
(1,50): error CS1056: Unexpected character ``
(1,52): error CS1002: ; expected
(1,52): error CS1056: Unexpected character ``
(1,64): error CS1525: Invalid expression term ')'
(1,66): error CS1002: ; expected
(1,75): error CS1733: Expected expression
```

csharp

```
public record Student(string FirstName, string LastName, int Grade);

var student = new Student("Ján", "Mrkvička", 2);
var (first, last, _) = student;
Console.WriteLine($"{first} {last}");
```

Ján Mrkvička

csharp

```
Zdroje
```

```
(1,1): error CS1024: Preprocessor directive expected
```

csharp

- [Records (C# reference)](<https://learn.microsoft.com/en-us/dotnet/csharp/language->
- [Introduction to record types in C#](<https://learn.microsoft.com/en-us/dotnet/csha>
- [What's new in C# 9.0 | Record types](<https://learn.microsoft.com/en-us/dotnet/csh>
- [Tutorials - Explore record types](<https://learn.microsoft.com/en-us/dotnet/csharp>
- [Mads Torgersen: C# 9.0 on the record](<https://devblogs.microsoft.com/dotnet/c-9-0>

(1,14): error CS1040: Preprocessor directives must appear as the first non-whitespace character on a line

(2,17): error CS1003: Syntax error, ',' expected

(2,20): error CS1003: Syntax error, ',' expected

(2,27): error CS1003: Syntax error, ',' expected

(2,33): error CS1003: Syntax error, ',' expected

(2,36): error CS1003: Syntax error, ',' expected

(2,37): error CS1040: Preprocessor directives must appear as the first non-whitespace character on a line

(3,8): error CS1003: Syntax error, ',' expected

(3,8): error CS1010: Newline in constant

(3,8): error CS1012: Too many characters in character literal

(4,24): error CS1003: Syntax error, ',' expected

(4,31): error CS1003: Syntax error, ',' expected

(5,9): error CS1003: Syntax error, ',' expected

(5,18): error CS1003: Syntax error, ',' expected

(5,20): error CS1003: Syntax error, ',' expected

(5,21): error CS1040: Preprocessor directives must appear as the first non-whitespace character on a line

(5,101): error CS1003: Syntax error, ']' expected

(5,101): error CS1026: ) expected

(5,101): error CS1003: Syntax error, ']' expected

(5,101): error CS1003: Syntax error, ']' expected

(5,101): error CS1026: ) expected

(5,101): error CS1003: Syntax error, ']' expected